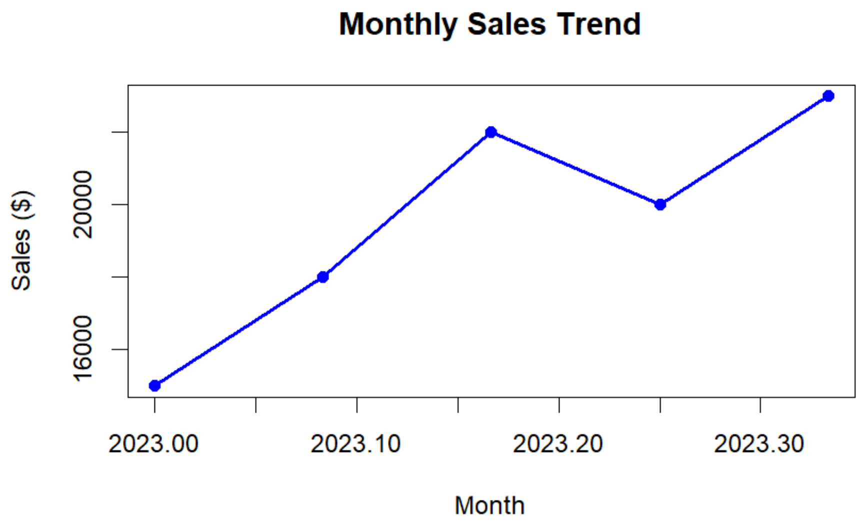
LAB EXPERIMENTS

SET-18

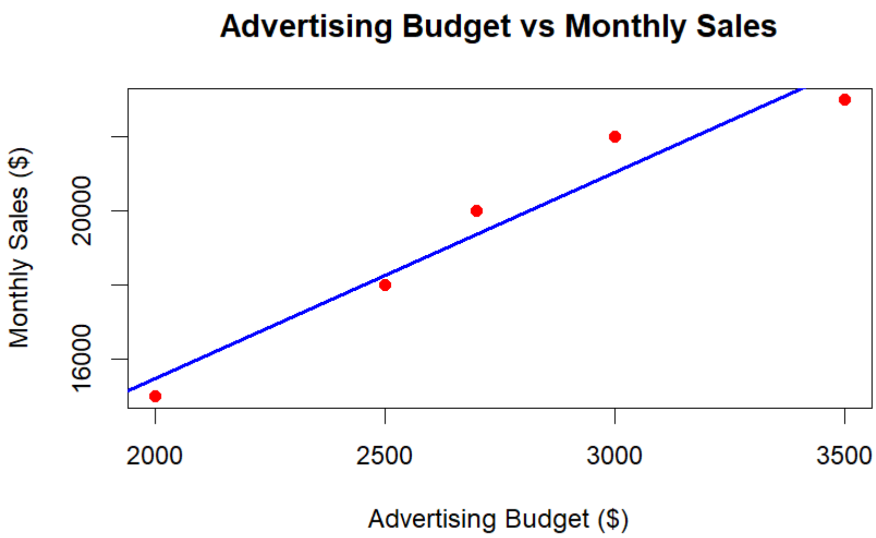
QUESTION-1



CODE:

```
# Monthly Sales Data
sales <- c(15000, 18000, 22000, 20000, 23000)
# Create time series
sales_ts <- ts(sales,
            start = c(2023, 1),
            frequency = 12)
# Plot time series
plot(sales_ts,
    type = "o",
    col = "blue",
    pch = 16,
    lwd = 2,
    xlab = "Month",
    ylab = "Sales ($)",
    main = "Monthly Sales Trend")
```

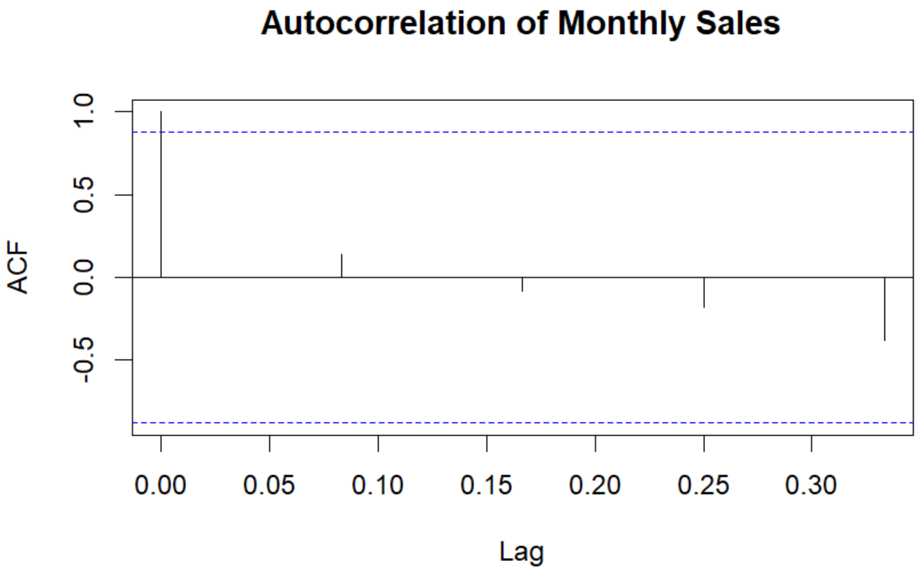
QUESTION-2



CODE:

```
# Advertising Budget
advertising <- c(2000, 2500, 3000, 2700, 3500)
# Monthly Sales
sales <- c(15000, 18000, 22000, 20000, 23000)
# Scatter Plot
plot(advertising,
     sales,
     pch = 19,
     col = "red",
     xlab = "Advertising Budget ($)",
     ylab = "Monthly Sales ($)",
     main = "Advertising Budget vs Monthly Sales")
# Regression Line
abline(lm(sales ~ advertising),
       col = "blue",
       lwd = 2)
```

QUESTION-3



CODE:

```
# Monthly Sales Data
sales <- c(15000, 18000, 22000, 20000, 23000)
# Create Time Series
sales_ts <- ts(sales,
               start = c(2023, 1),
               frequency = 12)
# Autocorrelation Plot
acf(sales_ts,
    main = "Autocorrelation of Monthly Sales")
```